

Lecture 3:

Probability Foundations & Bayes' Theorem

August 31, 2026

A random variable X is described by an ensemble:

$$(x, A_X, P_X)$$

outcome x , which takes on one of a set of possible values $A_X = \{a_1, a_2, \dots, a_n\}$
having probabilities $P_X = \{p_1, p_2, \dots, p_n\}$

$$p(x = a_i) = p_i \quad p_i \geq 0$$

$$\sum_{a_i \in A_X} p(x = a_i) = 1$$

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$$\sum_{a_i \in A_X} p(x = a_i) = 1$$

discrete

$$\int_{A_X} p(x) dx = 1$$

continuous

$p(x = a_i), p(a_i), p(x), P(X), Pr(X)$ all denote the probabilities

$p(X)$ = probability distribution of X

if X is continuous, **probability density function** (PDF)

we'll stick with $p(x)$ for simplicity

$x \sim p(x)$ — “ x is drawn from the distribution $p(x)$ ”

some more notations and rules of probability

$p(x, y)$ = **joint probability** of x and y

if T is a subset of A_X , $p(T) = p(x \in T) = \sum_{x \in T} p(x)$

marginal probability: $p(x) = \sum_{y \in A_Y} p(x, y) = \sum_y p(x, y)$

“marginalized over y ”

some more notations and rules of probability

$p(x, y)$ = **joint probability** of x and y

if T is a subset of A_X , $p(T) = p(x \in T) = \int_T p(x') \mathrm{d}x'$

marginal probability: $p(x) = \int_y p(x, y') \mathrm{d}y'$

“marginalized over y ”

some more notations and rules of probability

conditional probability: $p(x \mid y)$ — “probability of x given y ”

$$p(x \mid y) = \frac{p(x, y)}{p(y)} \text{ if } p(y) \neq 0$$

product rule (chain rule):

$$p(x, y) = p(x \mid y) p(y) = p(x) p(y \mid x)$$

$$p(x, y \mid z) = p(x \mid y, z) p(y \mid z) = p(x \mid z) p(y \mid x, z)$$

some more notations and rules of probability

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iff (if and only if) x and y are independent, $p(x \mid y) = p(x)$ and

$$p(x, y) = p(x) p(y)$$

some more notations and rules of probability

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product rule (chain rule):

$$p(x, y) = p(x \mid y) p(y) = p(x) p(y \mid x)$$

we can also rewrite the marginal as $p(x) = \int_y p(x, y') \, dy' = \int p(x \mid y') p(y') \, dy'$

Bayes' theorem

$$p(x, y) = p(x \mid y) p(y) = p(x) p(y \mid x)$$

Bayes' theorem

$$p(x, y) = p(x \mid y) p(y) = p(x) p(y \mid x)$$

$$p(y \mid x) = \frac{p(x \mid y) p(y)}{p(x)} = \frac{p(x \mid y) p(y)}{\int p(x \mid y) p(y) \, dy}$$

Bayes' theorem

$$p(x, y) = p(x | y) p(y) = p(x) p(y | x)$$

$$p(y | x) = \frac{p(x | y) p(y)}{p(x)} = \frac{p(x | y) p(y)}{\int p(x | y) p(y) dy}$$



Thomas Bayes, 1763

some more notations and rules of probability

$$1 = \int p(x) \, dx \quad \text{and} \quad 1 = \int p(x | y) \, dx \quad \text{but **cannot do** } \int p(x | y) \, dy$$

$$p(x | z) = \int p(x, y | z) \, dy$$

$$p(x | z) = \int p(x | y, z) p(y | z) \, dy$$

expectation value of a PDF $p(x)$

the probability-weighted average over every possible outcome:
the distribution's "center of mass"

$$\mathbb{E}[X] = \int x p(x) dx$$

expectation value of a PDF $p(x)$

the probability-weighted average over every possible outcome:
the distribution's "center of mass"

$$\mathbb{E}[X] = \int x \, p(x) \, dx$$

for any function f of X

$$\mathbb{E}[f(X)] = \int f(x) \, p(x) \, dx$$

“the law of the unconscious statistician”

A COVID test has a sensitivity of 84% and a true negative rate of 89%. You test positive. What's your chance of being infected? (*hint: Bayes theorem*)

the COVID rate in your area is 0.0025

A COVID test has a sensitivity of 84% and a true negative rate of 89%. You test positive. What's your chance of being infected? (*hint: Bayes theorem*)

$$p(+ | E) = 0.84$$

$$p(- | \tilde{E}) = 0.89$$

$$p(E) = 0.0025$$

$$p(E | +) = ?$$

A COVID test has a sensitivity of 84% and a true negative rate of 89%. You test positive. What's your chance of being infected? (*hint: Bayes theorem*)

$$p(+ | E) = 0.84; \quad p(- | \tilde{E}) = 0.89; \quad p(E) = 0.0025$$

$$p(E | +) = \frac{p(+ | E) p(E)}{p(+)} = \frac{p(+ | E) p(E)}{p(+ | E) p(E) + p(+ | \tilde{E}) p(\tilde{E})}$$

$$p(E | +) \approx 0.019$$

base rate fallacy (prior!)

A COVID test has a sensitivity of 84% and a true negative rate of 89%. You test positive. What's your chance of being infected? (*hint: Bayes theorem*)

$$p(+ | E) = 0.84; \quad p(- | \tilde{E}) = 0.89; \quad p(E) = 0.0025$$

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$$p(E | +) \approx 0.019$$

what happens if $p(E)$ increases to 0.5?

base rate fallacy (prior!)

some probability calculus exercises:

You have conditional PDFs $p(a | b)$, $p(a | d)$, $p(b | a, d)$, and $p(c | a, b, d)$.

Write expressions for $p(a, b | d)$, $p(b | d)$, and $p(a | c, d)$.

You have a function $g(b)$ and conditional PDFs $p(a | c)$ and $p(b | a, c)$.

What is the expectation value $\mathbb{E}(g | c)$?

some probability calculus exercises:

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Write expressions for $p(a, b | d)$, $p(b | d)$, and $p(a | c, d)$.

$$p(a, b | d) = p(b | a, d) p(a | d)$$
$$p(b | d) = \int p(b | a, d) p(a | d) \mathrm{d}a$$

some probability calculus exercises:

You have conditional PDFs $p(a | b)$, $p(a | d)$, $p(b | a, d)$, and $p(c | a, b, d)$.

Write expressions for $p(a, b | d)$, $p(b | d)$, and $p(a | c, d)$.

$$p(c | a, d) = \int p(c | a, b, d) p(b | a, d) db$$

$$p(a | c, d) = \frac{p(c | a, d) p(a | d)}{p(c | d)}, \quad p(c | d) = \int p(c | a, d) p(a | d) da$$

$$p(a | c, d) = \frac{p(a | d) \int p(c | a, b, d) p(b | a, d) db}{\iint p(c | a, b, d) p(b | a, d) p(a | d) db da}$$

some probability calculus exercises:

You have a function $g(b)$ and conditional PDFs $p(a \mid c)$ and $p(b \mid a, c)$.

What is the expectation value $\mathbb{E}(g \mid c)$?

$$\mathbb{E}(g \mid c) = \int g(b) p(b \mid c) db$$

$p(b \mid c)$ is not given but

$$p(b \mid c) = \int p(b \mid a, c) p(a \mid c) da$$

some probability calculus exercises:

You have a function $g(b)$ and conditional PDFs $p(a \mid c)$ and $p(b \mid a, c)$.

What is the expectation value $\mathbb{E}(g \mid c)$?

$$\begin{aligned}\mathbb{E}(g \mid c) &= \iint g(b) p(b \mid a, c) p(a \mid c) db da = \int p(a \mid c) \left[\int g(b) p(b \mid a, c) db \right] da = \int p(a \mid c) \mathbb{E}(g \mid a, c) da \\ &= \mathbb{E}_a \left[\mathbb{E}(g \mid a, c) \mid c \right]\end{aligned}$$

law of total expectation: $\mathbb{E}[X] = \mathbb{E} \left[\mathbb{E}[X \mid Y] \right]$

describing a distribution

$$\text{mean}(X) \equiv \mathbb{E}[X] = \int x p(x) \, dx$$

$$\text{var}(X) \equiv \mathbb{E}[(X - \mathbb{E}(X))^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

$$\text{std}(X) = \sqrt{\text{var}(X)}$$

common distributions in astro: **binomial distribution**

count of successes in a fixed number of independent yes/no trials, each with the same probability p (*transit detection*)

$$p(k | n, p) = \binom{n}{k} p^k (1 - p)^{n-k}$$

$$\text{mean} = np, \text{variance} = np(1 - p)$$

common distributions in astro: **Poisson distribution**

count of discrete, independent events in a fixed interval (time, area, solid angle)
at a constant average rate (*photon counts*)

$$p(k | \lambda) = \lambda^k \frac{e^{-\lambda}}{k!}$$

mean = variance = λ

photon counts: the dominant noise source for any photon-counting detector:
CCD pixels, X-ray and gamma-ray instruments

source counts: e.g., number of galaxies found in a patch of sky or volume
(massive end of the SMF)

common distributions in astro: **Gaussian distribution**

arises via the **Central Limit Theorem** whenever a measured quantity sums many small independent contributions

$$p(x | \mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

mean = μ , variance = σ^2

default noise model once you're not in the sparse-count regime.

statistical inference

We wanted to see everything our eyes would accommodate, to think what we could, and, out of our seeing and thinking, to build some structure in modeled imitation of observed reality.

The Log from the Sea of Cortez, *John Steinbeck*

so far we've been dealing with $p(x, y), p(x | y), p(y | x), p(y)...$

probabilistic/statistical inference:

if $x = \text{data}$ and $y = \theta$ (things we want to know)

what is statistical inference?

*some astronomical phenomenon
of interest (e.g. galaxies)*

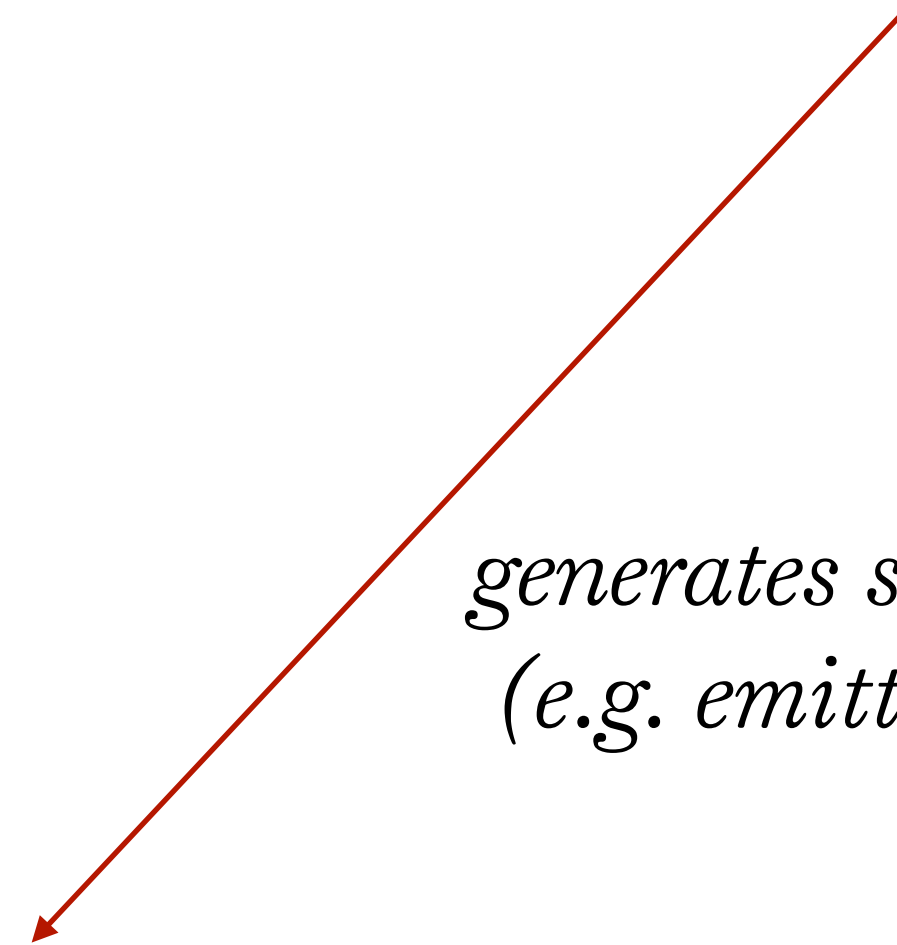


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*generates some data
(e.g. emitted light)*

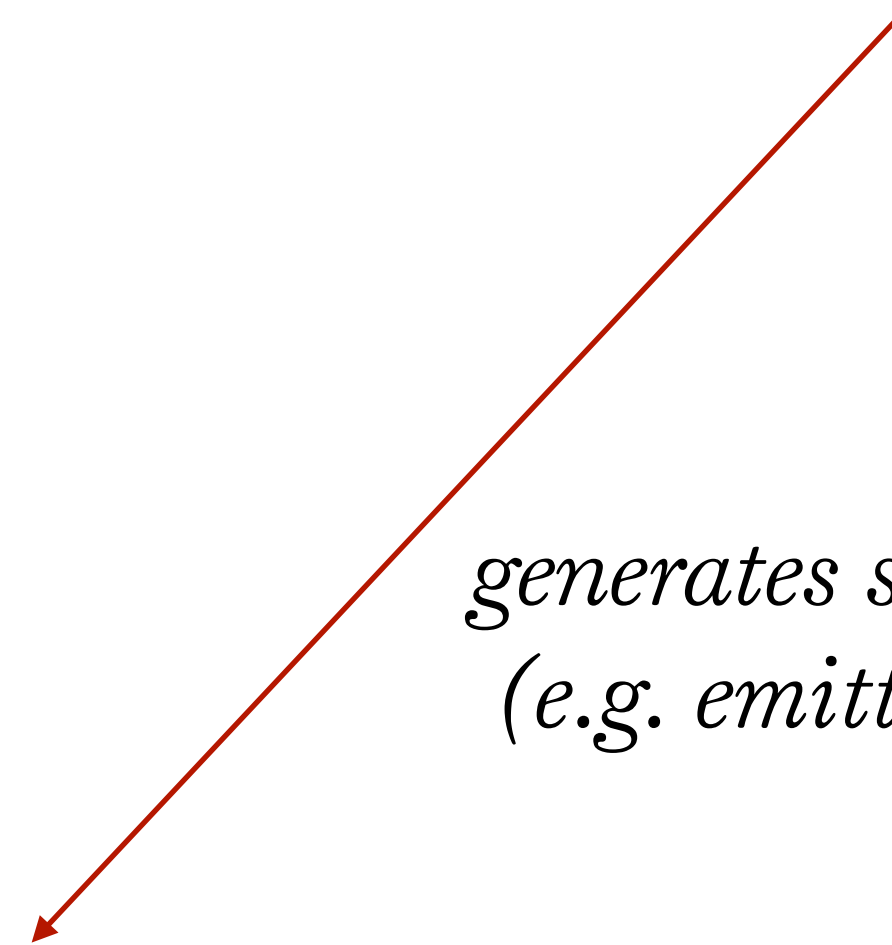


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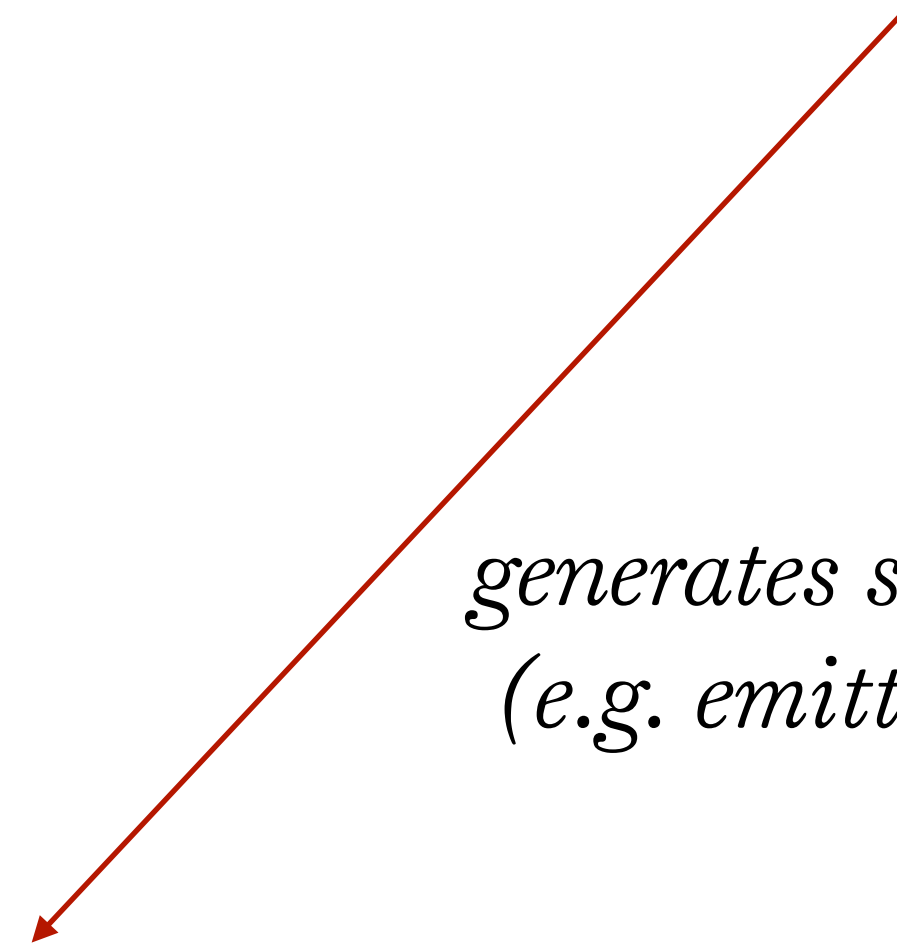
in an **ideal world** we would have access to all this data (*signal*) and
could use it to inform our physical model of the phenomenon

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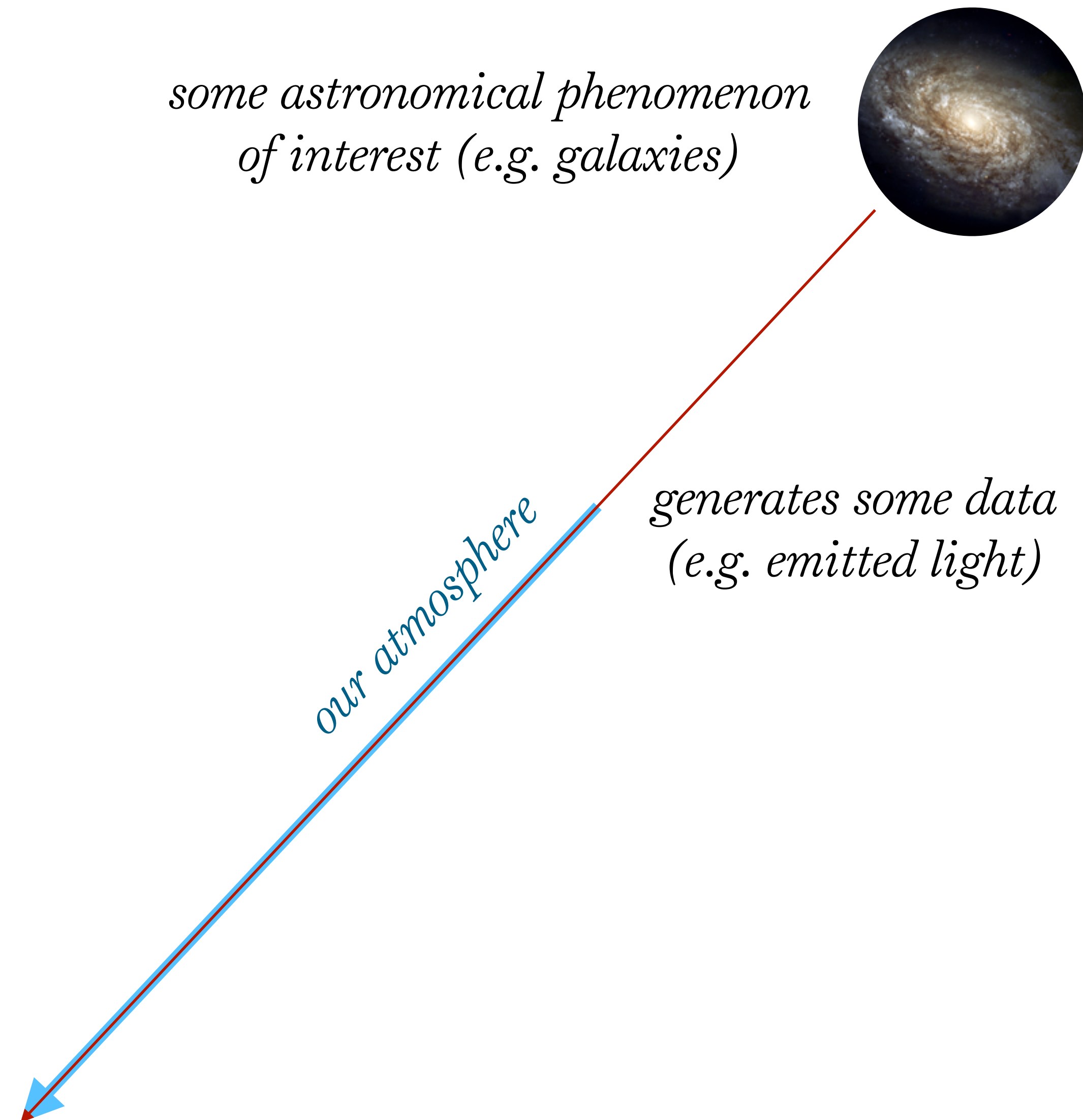
*generates some data
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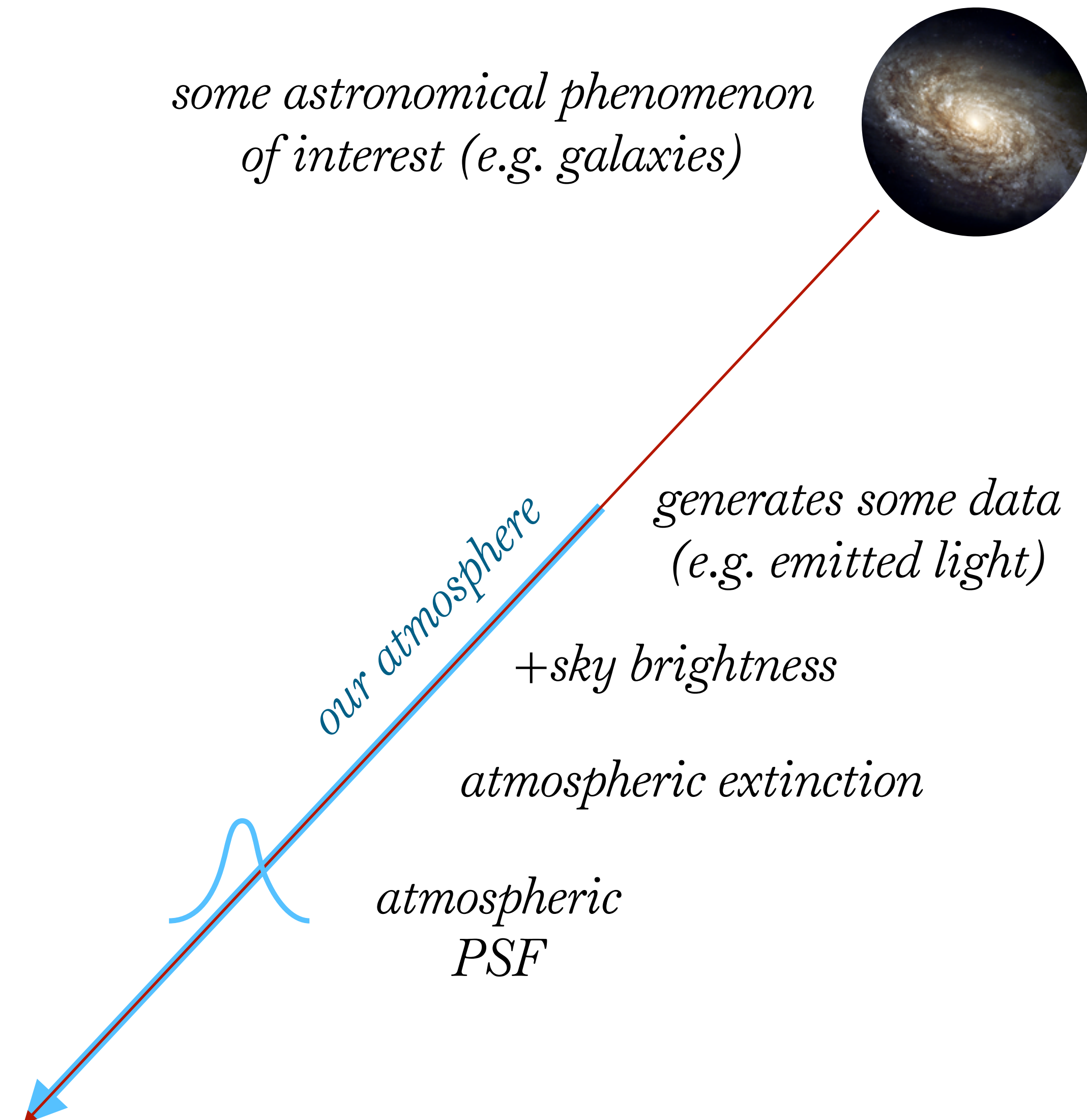
in an **ideal world** we would have access to all this data (*signal*) and
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but in the **real-world** we do not have access to all this data

what is statistical inference?

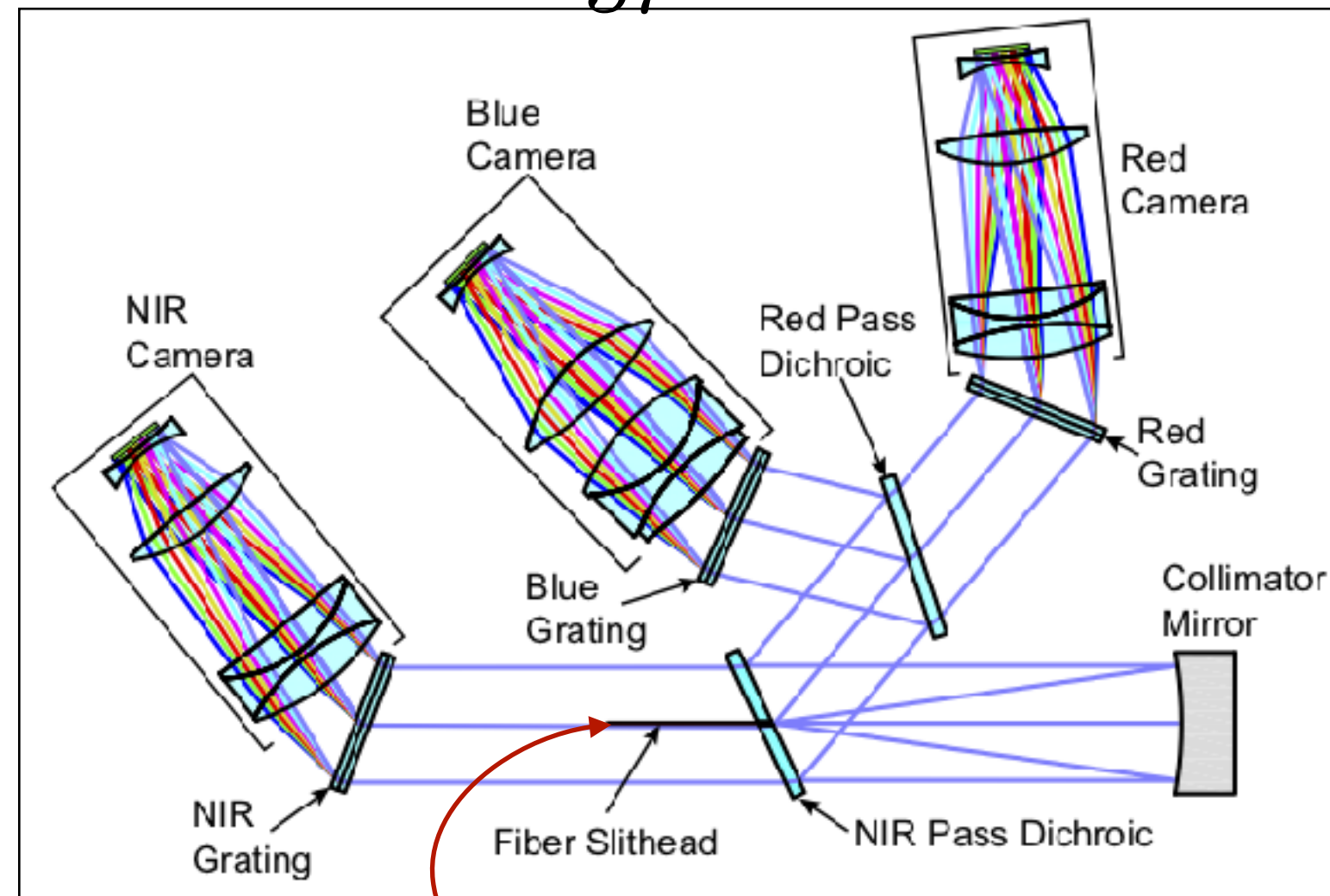


what is statistical inference?



what is statistical inference?

arXiv:1611.00037



*some astronomical phenomenon
of interest (e.g. galaxies)*



*generates some data
(e.g. emitted light)*

our atmosphere

+sky brightness

atmospheric extinction

*atmospheric
PSF*

optical fiber

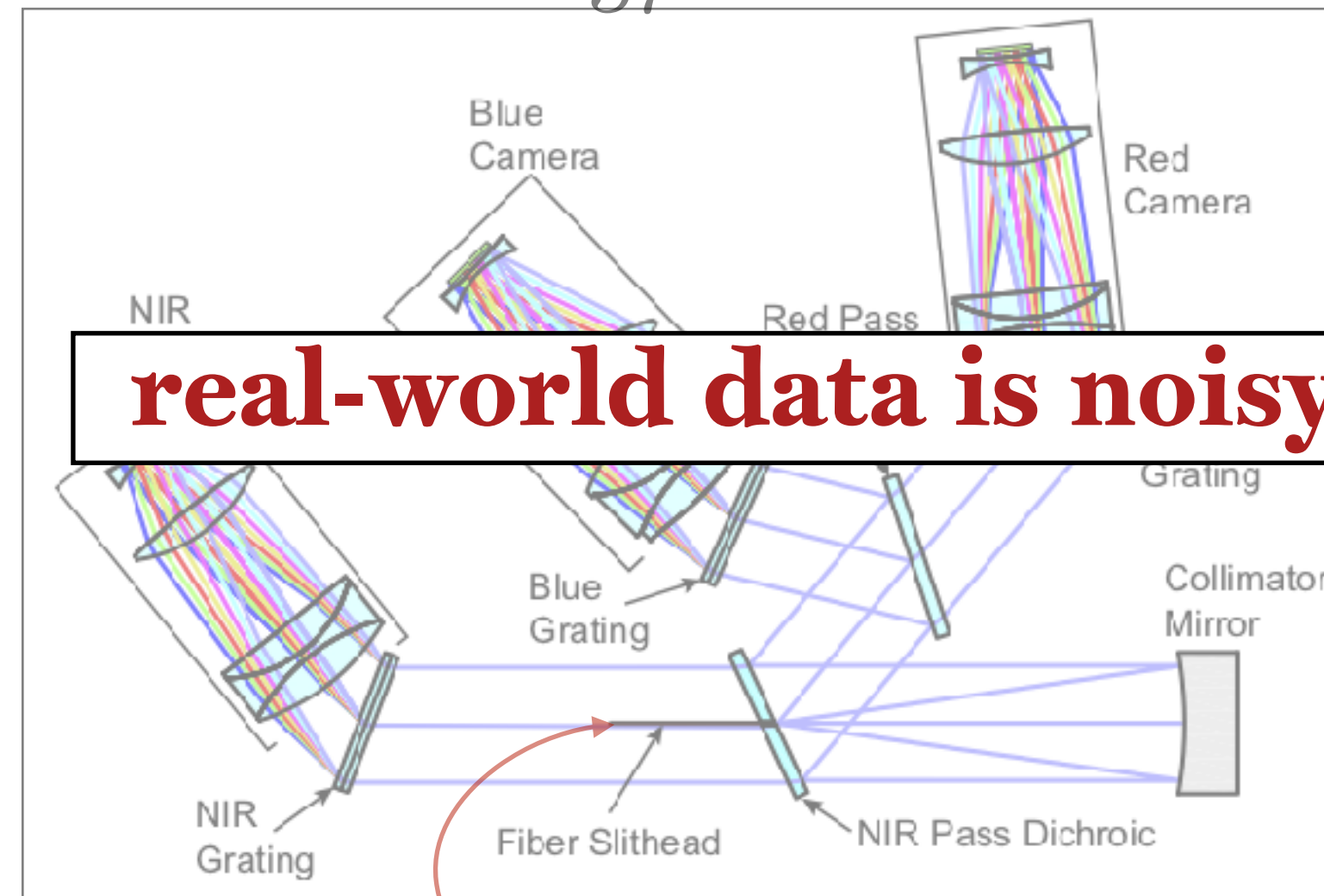
*telescope
PSF*

our instrument

primary mirror

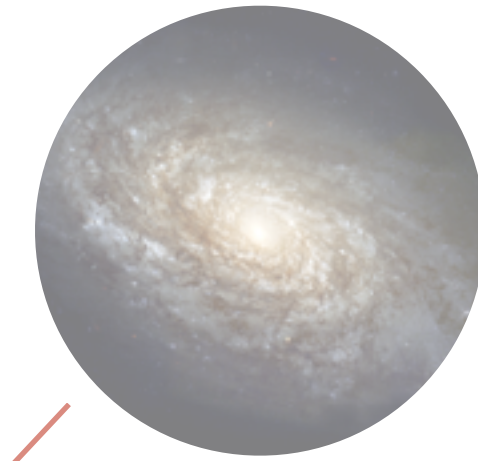
what is statistical inference?

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real-world data is noisy and incomplete

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of interest (e.g. galaxies)*



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(e.g. emitted light)*

+sky brightness

atmospheric extinction

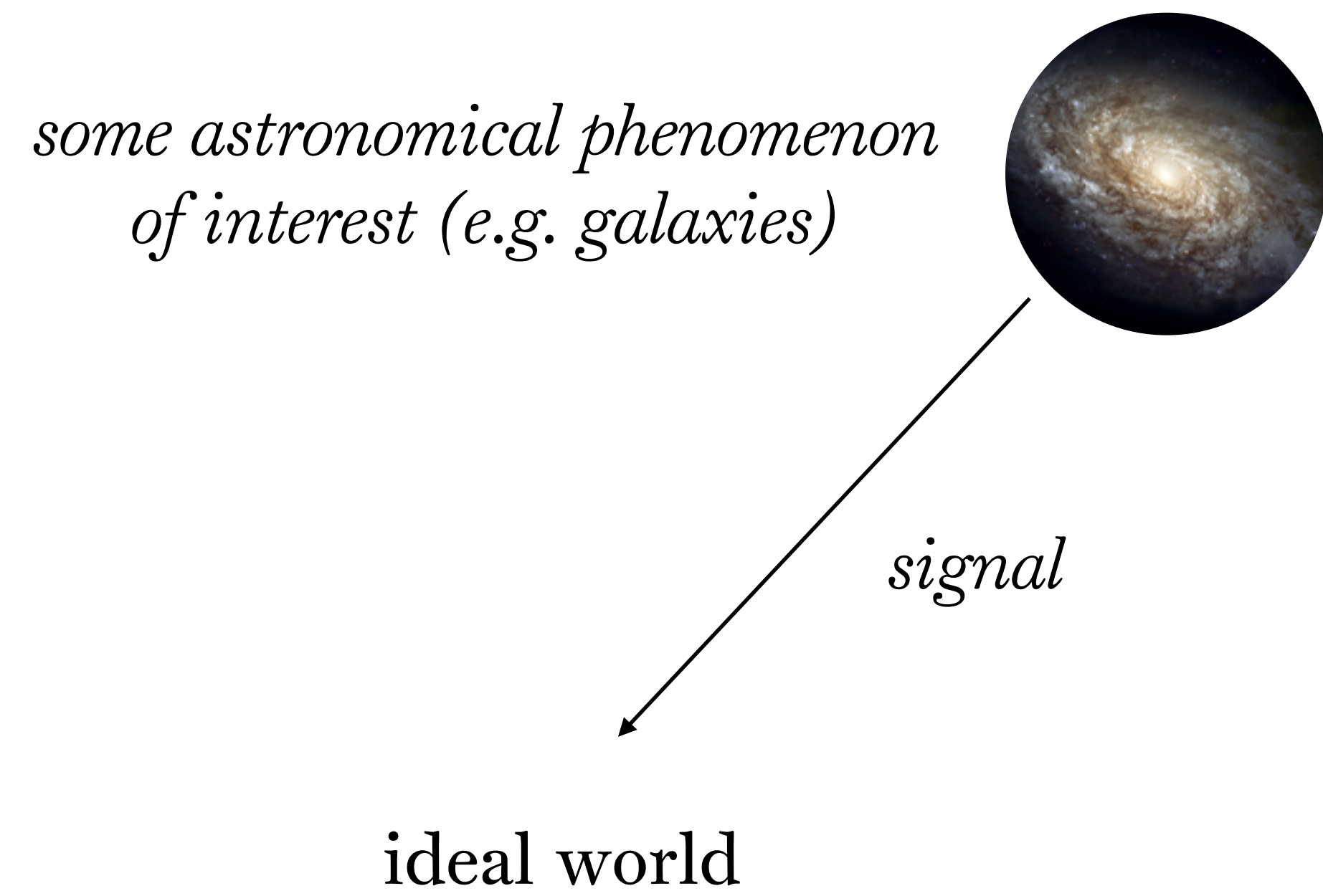
*atmospheric
PSF*

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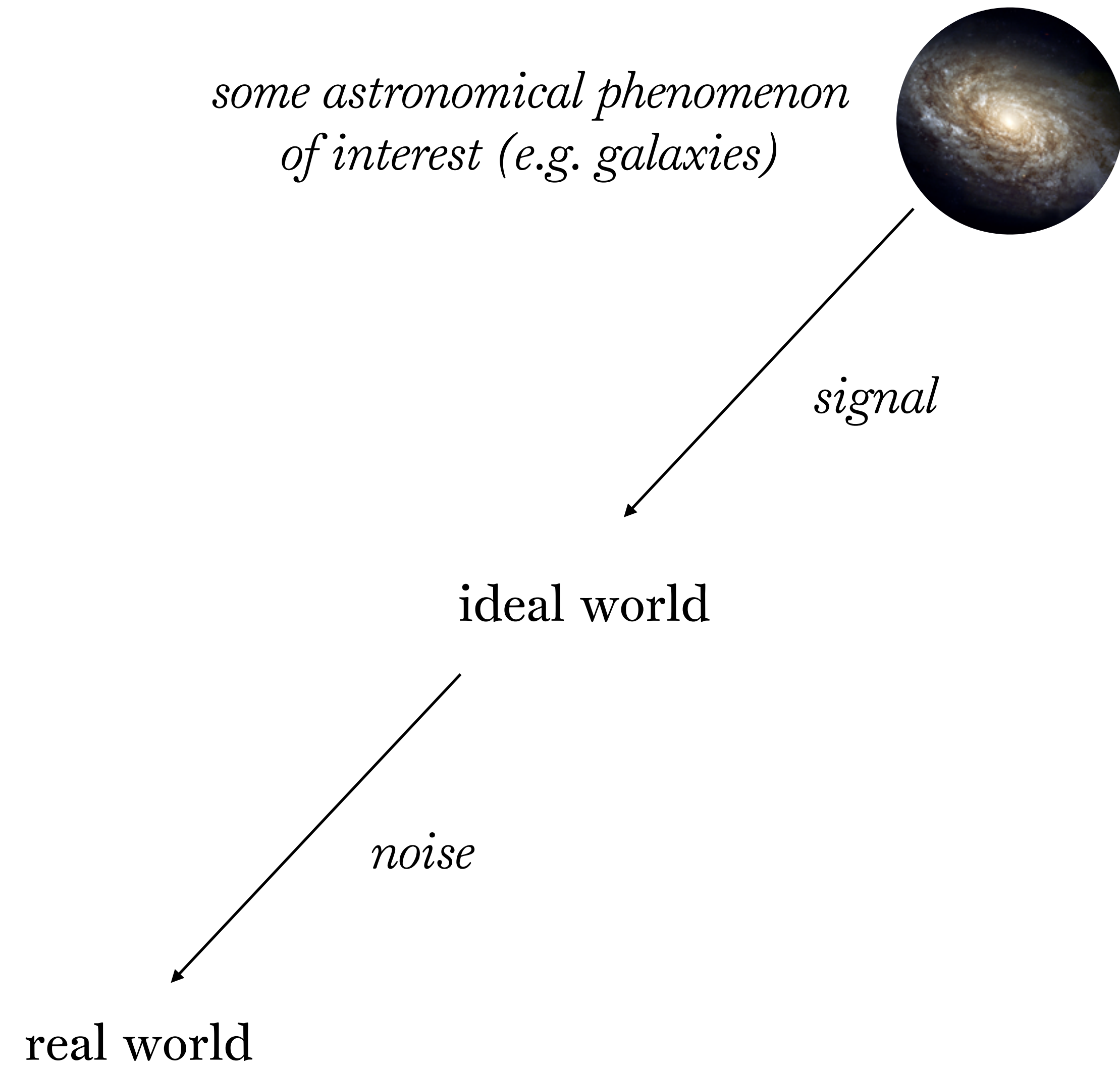
our instrument

primary mirror

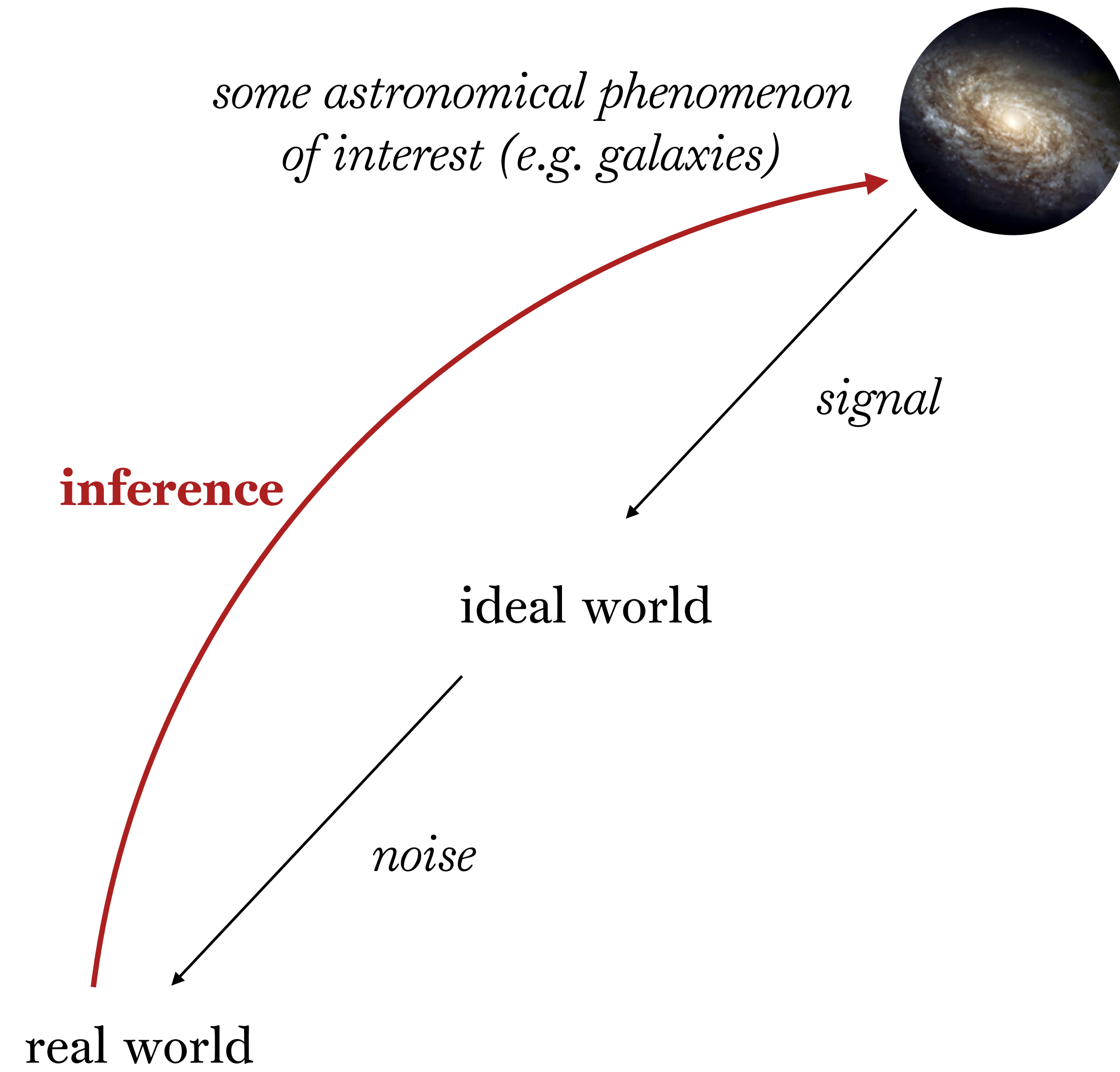
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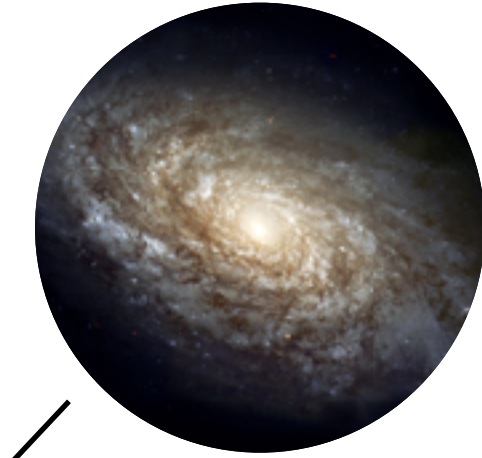


what is statistical inference? The framework that allows us to invert this process, converting data we collect to insights on the astrophysical phenomenon



e.g. inferring the physical properties of galaxies

*some astronomical phenomenon
of interest (e.g. galaxies)*



signal

ideal world

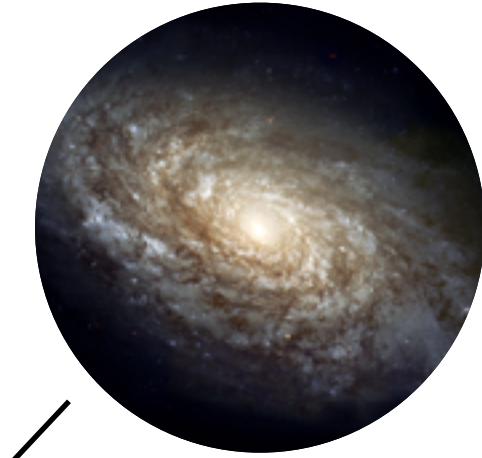
noise

real world

how much energy it emits across the entire electromagnetic spectrum

e.g. inferring the physical properties of galaxies

galaxy with some M_* , star formation and chemical enrichment history, etc.



signal

ideal world

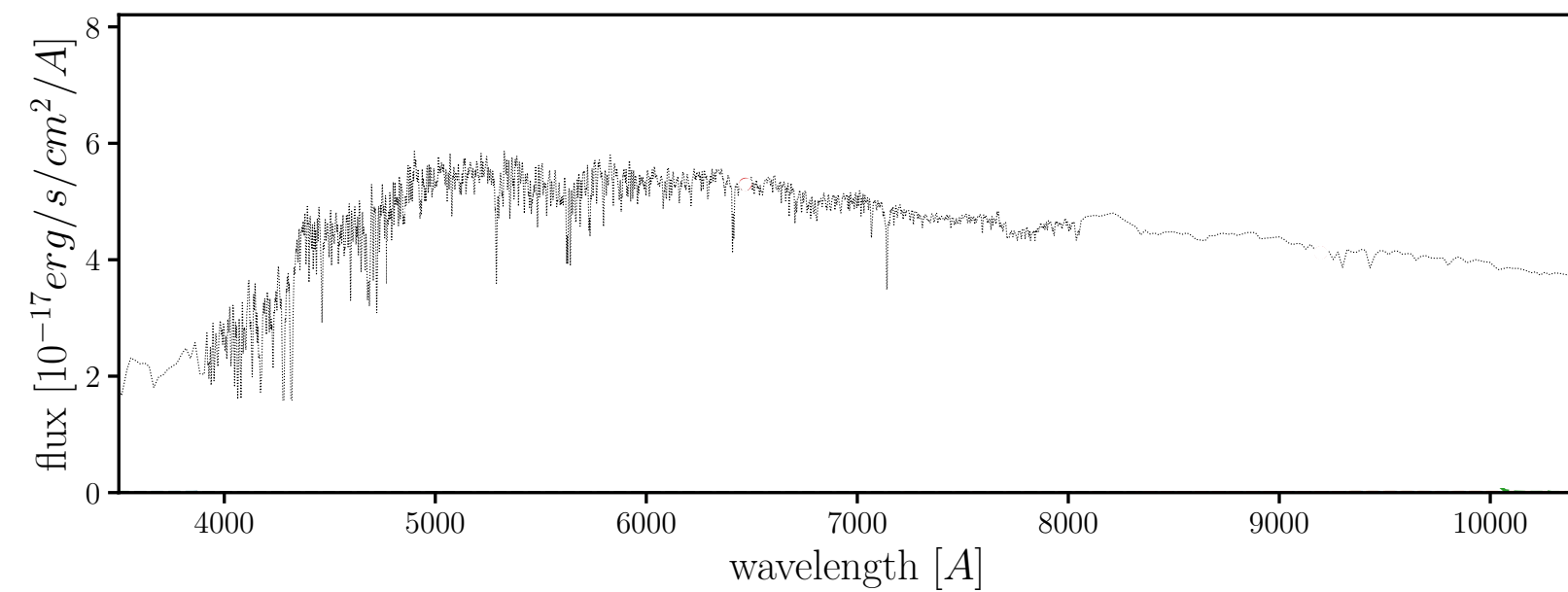
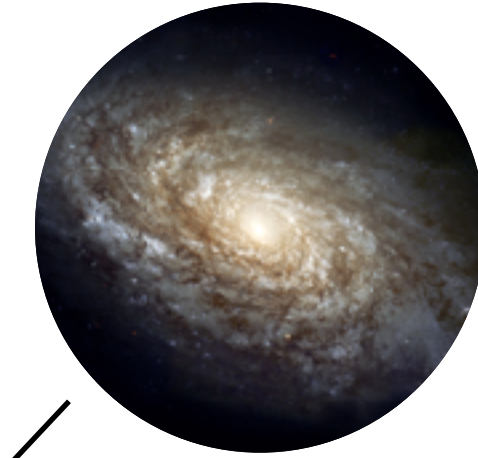
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**spectral energy distribution
(SED)**

ideal world

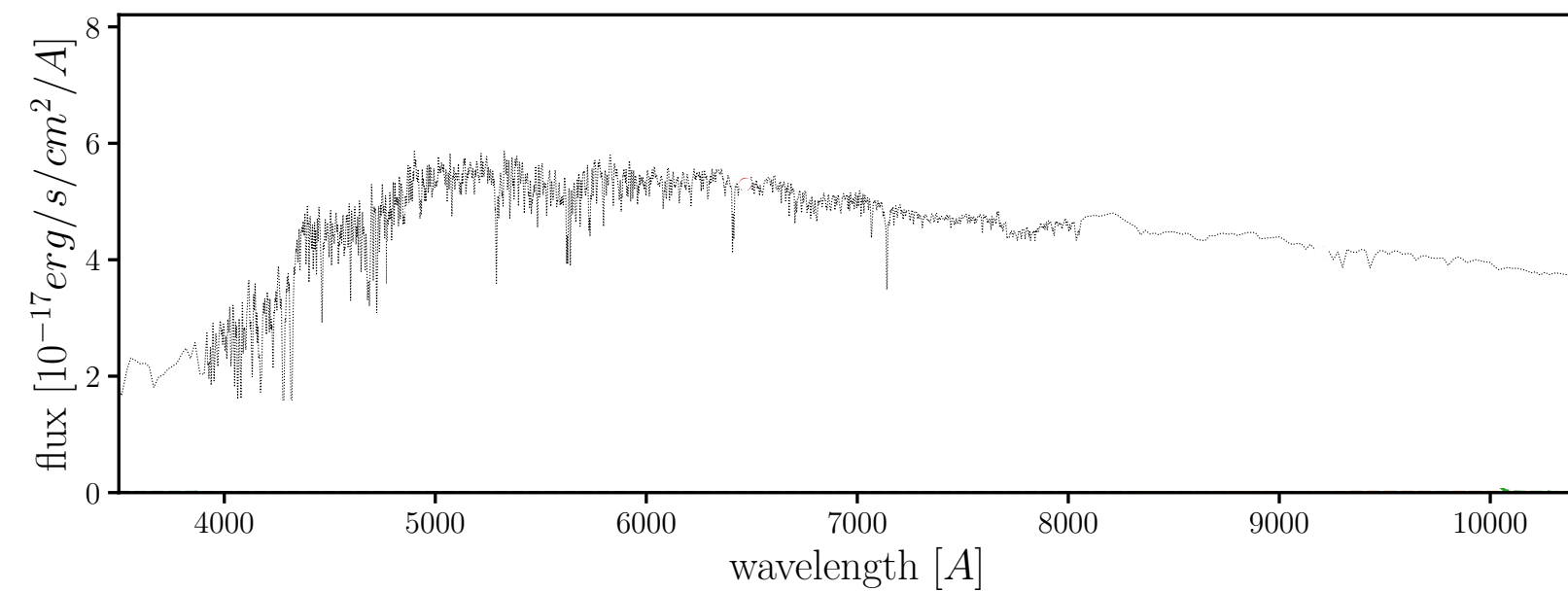
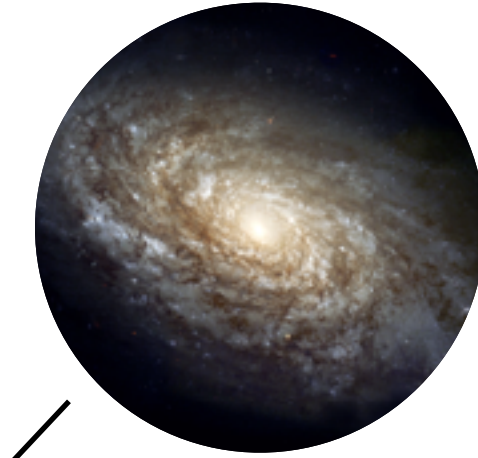
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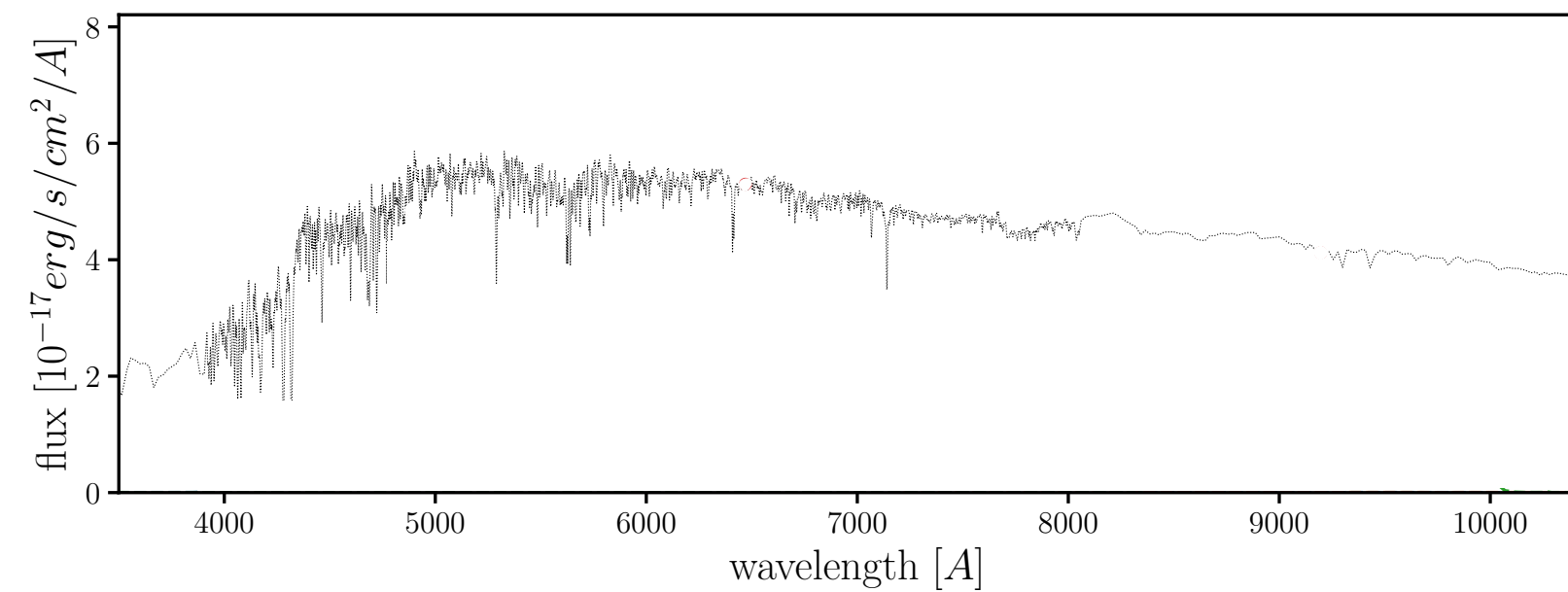
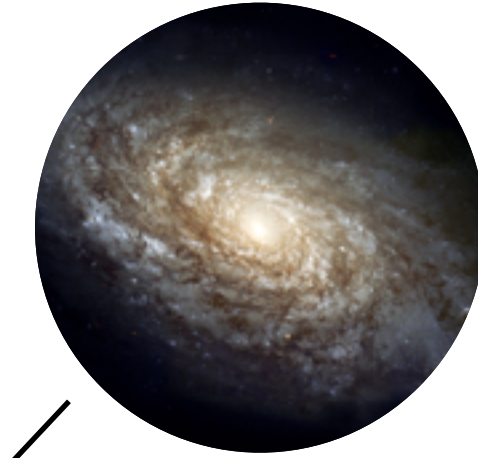
ideal world

**noise from ground based optical telescope
(e.g. Dark Energy Spectroscopic Instrument)**

real world

e.g. inferring the physical properties of galaxies

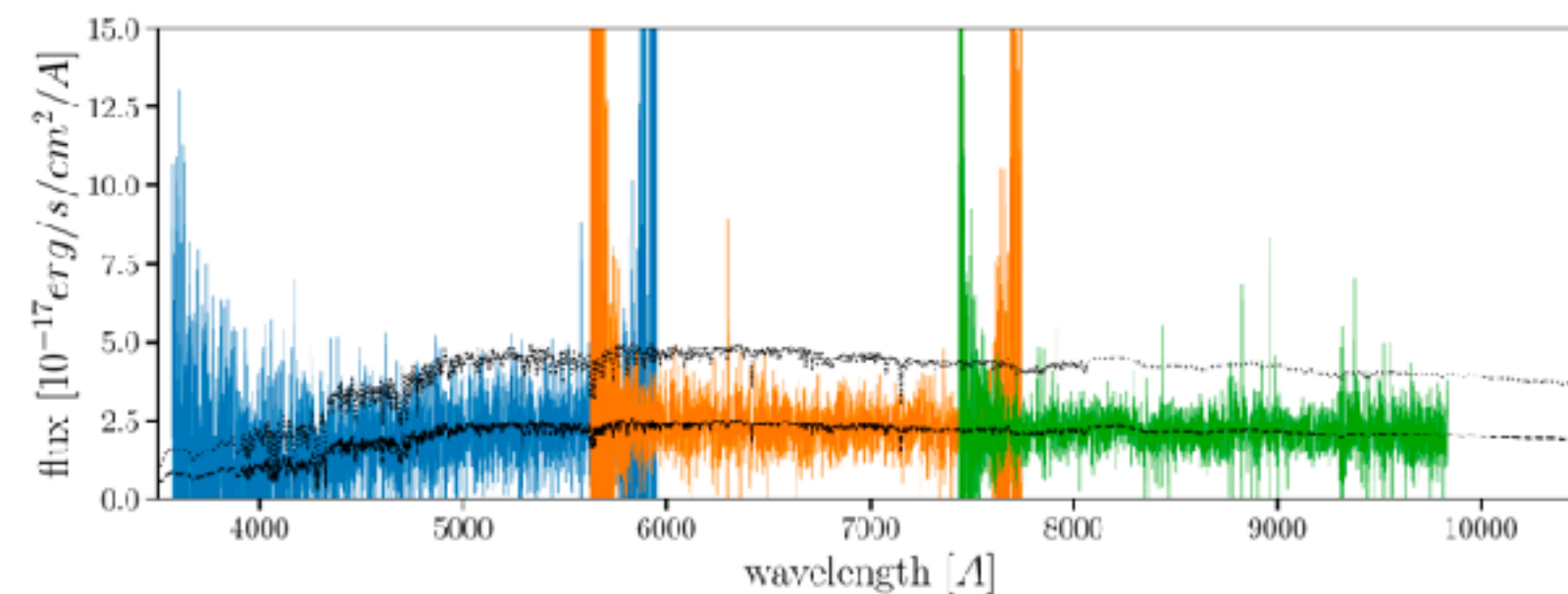
galaxy with some M_* , star formation and chemical enrichment history, etc.



spectral energy distribution (SED)

ideal world

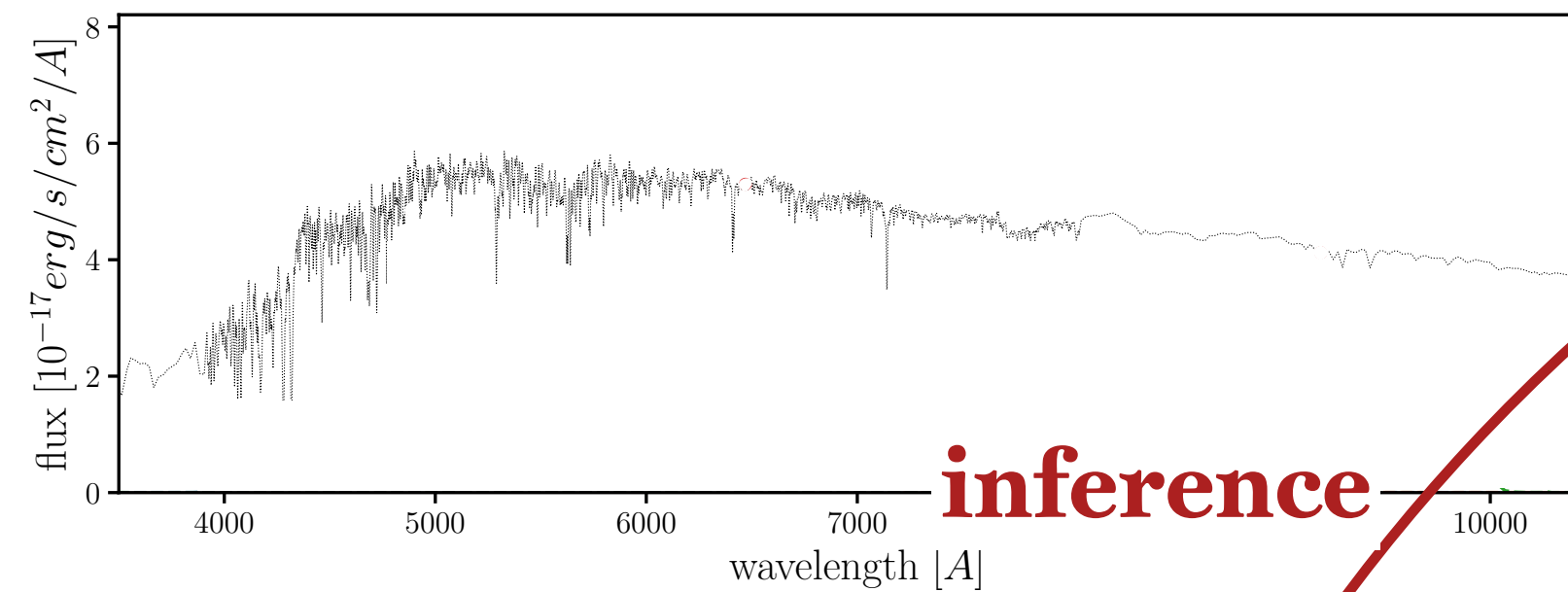
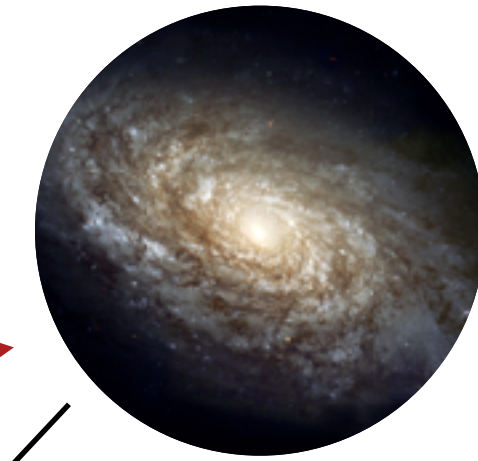
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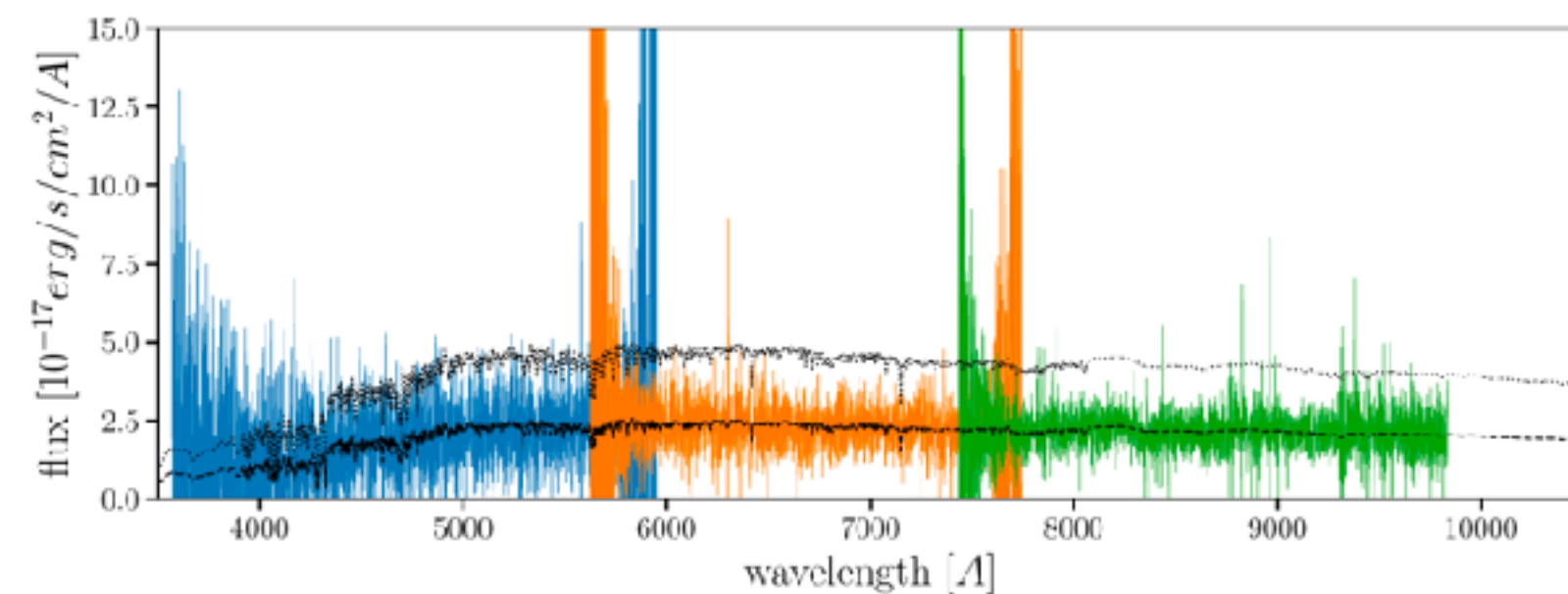
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spectral energy distribution (SED)

ideal world

noise from ground based optical telescope
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real world

two inference frameworks: **Frequentist** vs **Bayesian**

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fundamental difference: *how you think about probabilities*

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fundamental difference: *how you think about probabilities*

Frequentist: *frequencies of real or hypothetical events*

two inference frameworks: **Frequentist** vs **Bayesian**

fundamental difference: *how you think about probabilities*

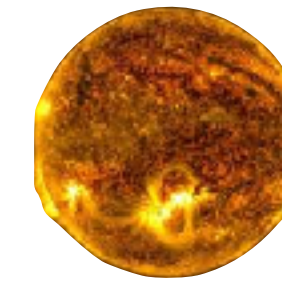
Frequentist: *frequencies of real or hypothetical events*

Bayesian: *degree of knowledge*

two inference frameworks: **Frequentist** vs **Bayesian**

star with true flux

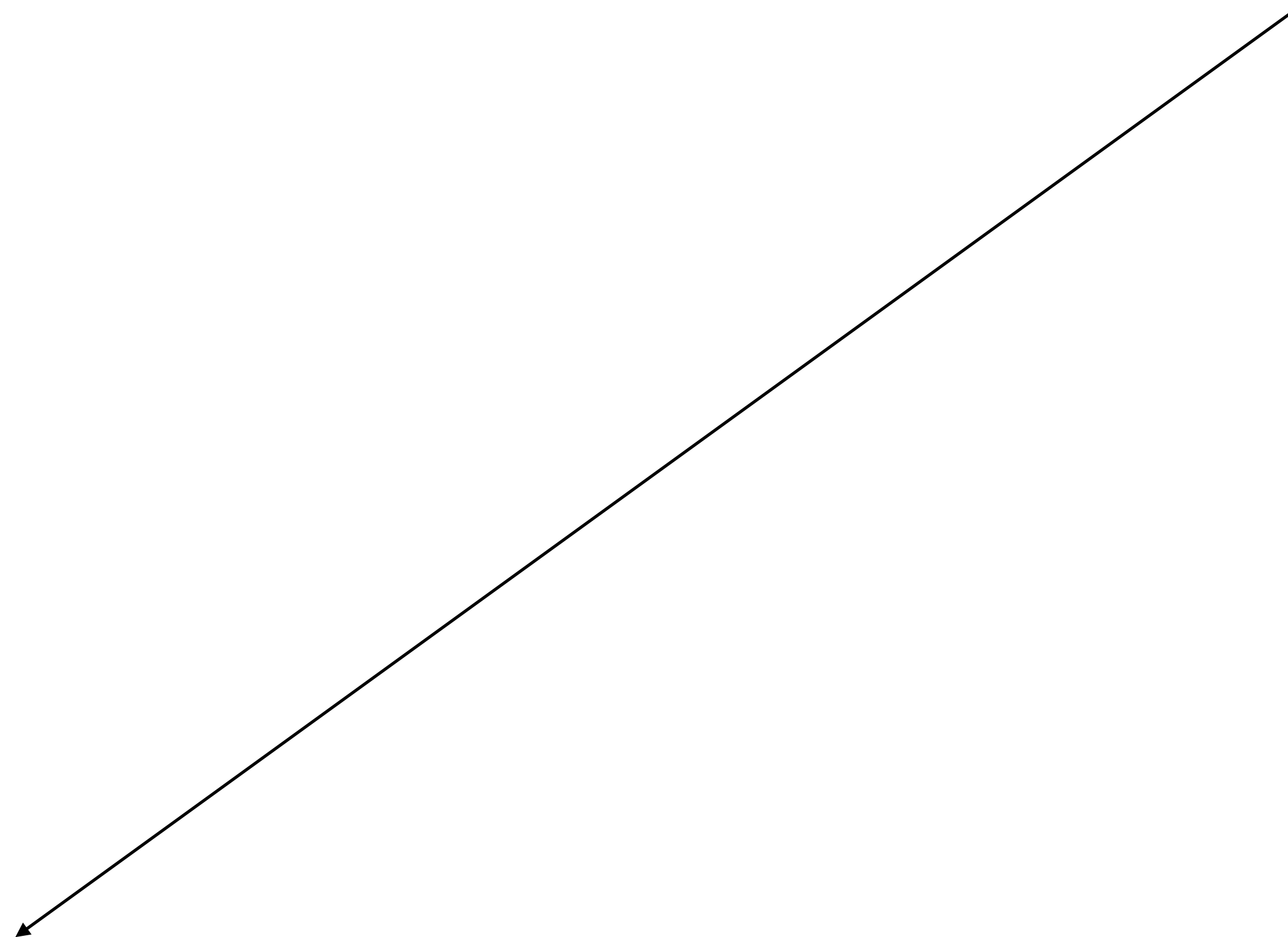
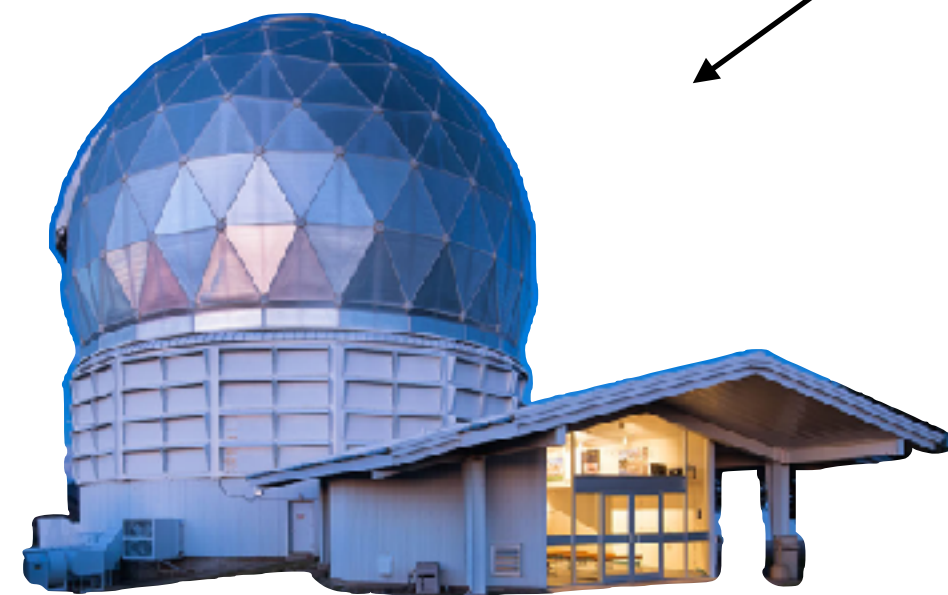
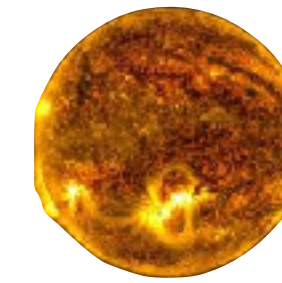
F_{true}



two inference frameworks: **Frequentist** vs **Bayesian**

star with true flux

F_{true}

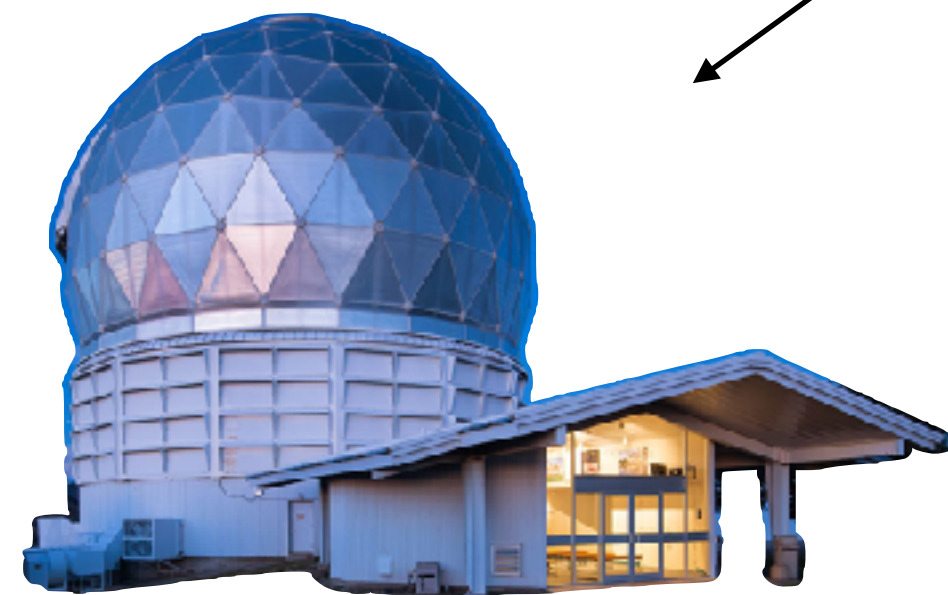
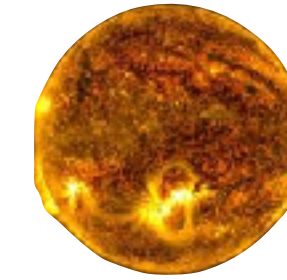


hypothetical example

two inference frameworks: **Frequentist** vs **Bayesian**

star with true flux

F_{true}



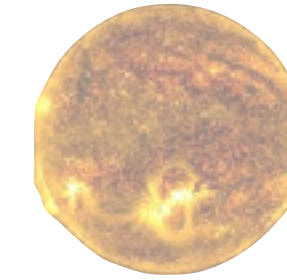
N measurements:
 F_i with uncertainty e_i

hypothetical example

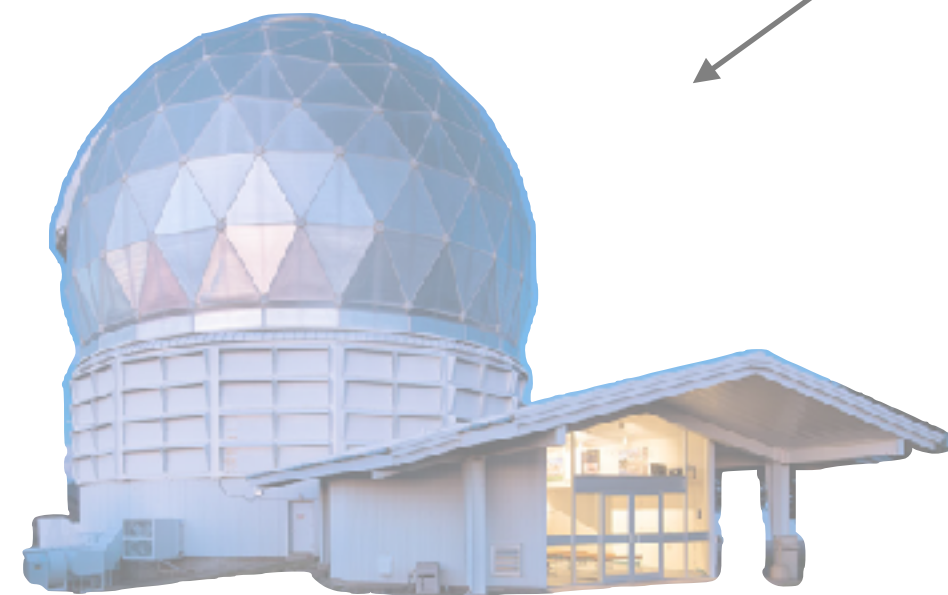
two inference frameworks: **Frequentist** vs **Bayesian**

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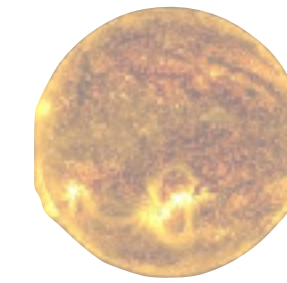
Frequentist: Each measure F_i is slightly different due to the statistical errors of the measuring process. In the limit of a large number of measurements, the frequency of any given value indicates the probability of measuring that value.



N measurements:
 F_i with uncertainty e_i

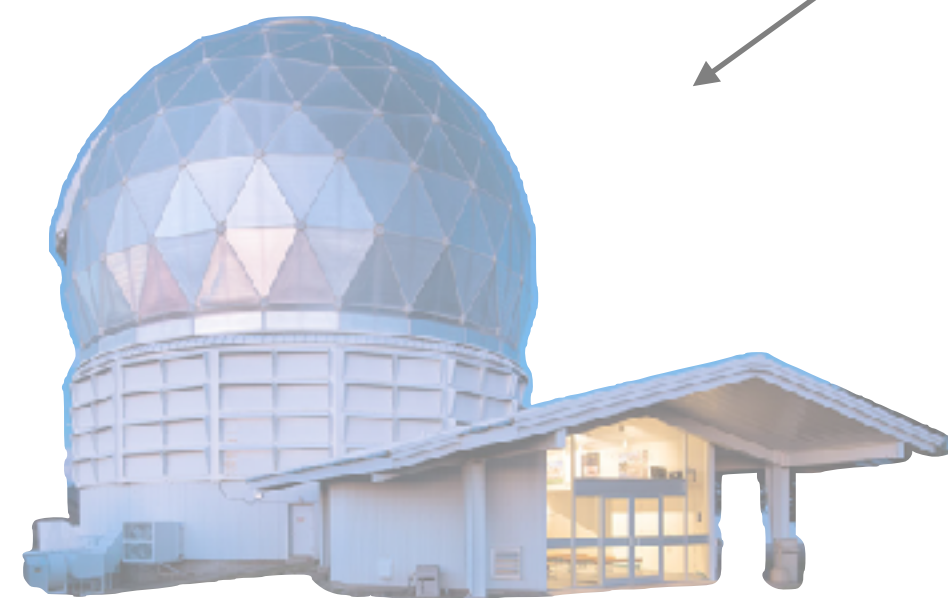
two inference frameworks: **Frequentist** vs **Bayesian**

star with true flux
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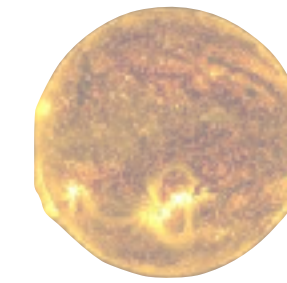
Bayesian: We measure the flux of star with some probability $p(F)$. Even if we believe that the star has some true fixed flux F_{true} , we only have a finite certainty about it based on prior information and/or available data.



N measurements:
 F_i with uncertainty e_i

two inference frameworks: **Frequentist** vs **Bayesian**

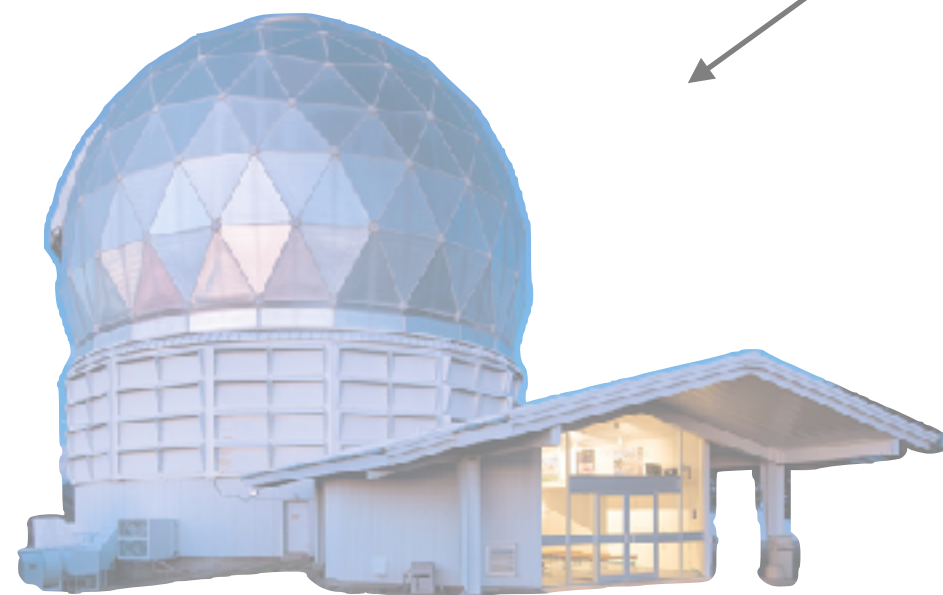
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 F_{true}



Frequentist: Each measure F_i is slightly different due to the statistical errors of the measuring process. In the limit of a large number of measurements, the frequency of any given value indicates the probability of measuring that value.

Bayesian: We measure the flux of star with some probability $p(F)$. Even if we believe that the star has some true fixed flux F_{true} , we only have a finite certainty about it based on prior information and/or available data.

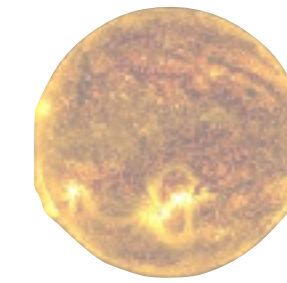
$p(F)$ can be estimated from frequencies in a large number of measurements, but this is not fundamental



N measurements:
 F_i with uncertainty e_i

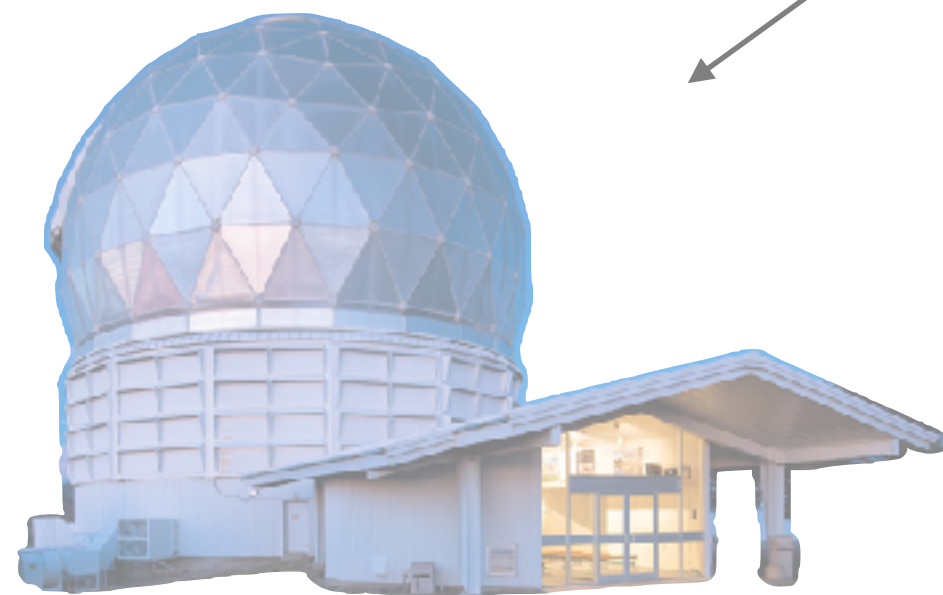
two inference frameworks: **Frequentist** vs **Bayesian**

star with true flux
 F_{true}



Frequentist: Each measure F_i is slightly different due to the statistical errors of the measuring process. In the limit of a large number of measurements, the frequency of any given value indicates the probability of measuring that value.

Bayesian: We measure the flux of star with some probability $p(F)$. Even if we believe that the star has some true fixed flux F_{true} , we only have a finite certainty about it based on **prior** information and/or available data.



N measurements:
 F_i with uncertainty e_i

priors in Bayesian inference

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Bayes' Theorem

$$p(\theta | X) = \frac{p(\theta) p(X | \theta)}{p(X)}$$

θ : parameters of your model

X : data

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Frequentist: The prior is problematic when no true prior information is available and its subjectivity has no place in statistical data analysis

Bayesian: Frequentists don't solve this problem! The Frequentist approach can be viewed as simply a special case of the Bayesian approach for some (implicit) choice of the prior. It's better to make this implicit choice explicit

we'll mainly focus on the Bayesian approach

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In astronomy, we can't rerun the universe to get more data.

You only observe one sky.

Let's implement Frequentist and Bayesian inference!